VISION-BASED ROBOTIC ARM FOR SORTING STATIONARY OBJECTS BASED ON COLOR, SHAPE, AND SIZE

PROJECT REPORT

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Abstract

This project presents a vision-based robotic arm system capable of sorting stationary objects based on color and shape using computer vision techniques. The system uses a vision sensor to capture images of objects placed on a conveyor belt. Image processing techniques implemented using OpenCV are applied to detect object color and shape. After detection, the robotic arm performs pick-and-place operations to move the objects to designated locations.

The simulation is implemented in CoppeliaSim using a Dobot robotic arm model. A proximity sensor detects the arrival of an object on the conveyor belt. Once detected, the conveyor stops and the vision sensor captures an image. The system then identifies the object's color and shape using HSV color filtering and contour analysis. Based on the detected object features, the robotic arm moves using inverse kinematics to pick the object and place it at the sorting location.

Introduction

Automation using robotics and computer vision has become increasingly important in industrial environments. Robotic arms are widely used for repetitive tasks such as assembly, sorting, packaging, and material handling. Computer vision enables robots to analyze images and make intelligent decisions based on visual information.

This project focuses on developing a vision-based robotic arm system capable of detecting and sorting stationary objects based on visual features such as color and shape. The system integrates a vision sensor, image processing algorithms, and robotic arm control to perform automated sorting.

Problem Statement

Manual sorting of objects is inefficient and prone to human error. Industries require automated systems that can quickly detect and classify objects. Therefore, this project aims to design a system capable of detecting object features using computer vision and automatically sorting them using a robotic arm.

Objectives

- Capture images using a vision sensor.
- Detect object color using HSV color segmentation.
- Detect object shape using contour detection.
- Control a robotic arm to perform pick-and-place operations.
- Demonstrate automated sorting in a simulation environment.

Literature Review

Computer vision and robotics integration has been widely studied in industrial automation. Many modern manufacturing systems use robotic arms combined with cameras and sensors for object detection and manipulation. Techniques such as color segmentation, contour detection, and shape approximation are commonly used for object classification.

OpenCV provides efficient tools for image processing and has been widely used in robotics applications. This project applies these techniques to create a simple and effective sorting system.

System Architecture

The system architecture consists of the following components:

- Vision Sensor (Camera)
- Image Processing Module
- Object Detection Module
- Robotic Arm Controller
- Conveyor Belt System
- Sorting Area

Architecture Flow:

Camera → Image Processing (OpenCV) → Feature Detection → Decision System → Robotic Arm → Sorting Area

Methodology

The methodology implemented in the code follows these steps:

1. The simulation starts and gravity is initialized.
2. The conveyor belt begins moving objects.
3. A proximity sensor detects when an object arrives.
4. Once detected, the conveyor belt stops.
5. The vision sensor captures an image of the object.
6. The image is converted from RGB to BGR and then to HSV color space.
7. HSV thresholds are applied to detect red and blue colored objects.
8. Contours are extracted from the binary masks.
9. Small contours (area < 1500) are ignored.
10. Shape detection is performed using contour approximation:
    - 3 vertices → Triangle
    - 4 vertices → Square
    - More than 6 vertices → Circle
11. After detection, inverse kinematics is used to calculate robotic arm joint positions.
12. The robotic arm moves to the object location.
13. The suction cup picks the object.
14. The arm moves to the target sorting location.
15. The object is released and the conveyor restarts.

System Flowchart

Start

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Initialize Simulation

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Start Conveyor

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Proximity Sensor Detects Object

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Stop Conveyor

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Capture Image from Vision Sensor

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Convert Image to HSV

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Detect Color

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Detect Shape using Contours

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Classify Object

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Calculate IK for Robot

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Move Robot to Object

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Pick Object

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Move to Target Location

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Place Object

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Restart Conveyor

↓

End

Image Processing Workflow

1. Capture image from vision sensor
2. Convert image to HSV color space
3. Apply color threshold for red and blue
4. Apply median blur to remove noise
5. Detect contours
6. Calculate contour area
7. Approximate polygon to detect shape
8. Classify object

Hardware Requirements

- Dobot Robotic Arm
- Vision Sensor / Camera
- Computer System
- Conveyor Mechanism

Software Requirements

- Python Programming Language
- OpenCV Library
- NumPy
- CoppeliaSim Simulation Software
- Remote API Client

Implementation

The simulation is implemented in CoppeliaSim using Python and OpenCV. The program connects to the simulator using the Remote API client. The conveyor moves objects toward a proximity sensor. When an object is detected, the conveyor stops and an image is captured from the vision sensor.

The image processing module identifies the object color and shape using HSV segmentation and contour detection. After classification, inverse kinematics is used to compute joint configurations for the robotic arm to pick and place the object.

Code Implementation

The program uses OpenCV for vision detection and CoppeliaSim remote API for robot control. Key tasks performed in the code include:

- Capturing images from the vision sensor
- Converting images from RGB to BGR and HSV
- Detecting colors using HSV thresholds
- Detecting contours and identifying shapes
- Controlling robot joints using inverse kinematics

Results

The simulation successfully detected red and blue objects and classified shapes such as triangles, squares, and circles. The robotic arm successfully picked the objects using the suction cup and placed them in the designated location.

The system demonstrated reliable performance within the simulation environment.

Advantages

- Reduces manual labor
- Improves sorting accuracy
- Demonstrates integration of robotics and computer vision
- Cost-effective automation solution

Applications

- Industrial automation
- Packaging systems
- Warehouse sorting
- Robotics education laboratories

Conclusion

The project successfully demonstrates a vision-based robotic sorting system using computer vision and robotic manipulation. The integration of OpenCV image processing and robotic arm control enables automated object sorting based on visual features.

Future Scope

- Implement deep learning-based object detection
- Add more colors and shapes
- Integrate real hardware robotic arms
- Implement real-time conveyor sorting

References

1. OpenCV Documentation
2. CoppeliaSim User Manual
3. Computer Vision: Algorithms and Applications – Richard Szeliski